Fiscal Year:	FY 2010	**Task Last Updated:**	FY 09/30/2010
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Project Title:	Heat Transfer on Multiscale Textures with Electrowetting Control		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	91360-2362	**Congressional District:**	24
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	07/01/2009	**End Date:**	06/30/2012
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Cheng, Jiangtao (Teledyne Scientific Company)		
Grant/Contract No.:	NNC09CA36C		
Performance Goal No.:			
Performance Goal Text:			
	Modern electronic systems used in space must be reliable and efficient with thermal management unaffected by outer space environments. Heat pipes and vapor chambers based on phase-change heat transfer mechanisms have proven to be an efficient approach to dissipate heat in many space or ground-based thermal-energy systems. However, a layer of capillary wick structure fully covering the chamber walls, e.g., condenser and evaporator surfaces, creates additional thermal resistance, because the effective thermal conductivity is much lower in saturated wicks. This thermal barrier can generate an unacceptable temperature rise, especially in high heat-flux applications. To overcome this issue, we propose to develop an innovative wickless and electrowetting-controlled vapor chamber with bioinspired two-tier textures on the chamber wall. These engineered superhydrophobic / superhydrophilic multiscale textures, formed by growing carbon nanotubes (CNTs) on top of micropillars on a silicon chip, mimic the structure of lotus leaves; they can promote dropwise condensation at the condenser site, and enhance film evaporation at the evaporator site. The condensate droplets will be dislodged from the condenser site and circulated to the heat source		

Task Description:	via electrowetting transport. Droplet transport by electrowetting without any mechanical moving parts has great potential in thermal management due to its programmable and reconfigurable characteristics, prompt response, and low power consumption. Most importantly, given a two-dimensional array of control electrodes, condensate can be transported over the low-friction hydrophobic surface to a hot spot without the significant pressure loss that is associated with fine pore wicks. The transported droplets will be captured by a few finger-like extensions from the heat source, which are hydrophilized by a coating of UV&Ozone-treated carbon nanotubes. This hydrophilic coating will force droplets to cling to the hot spot surface, changing heat transfer into the more efficient form of thin film evaporation. In this program, we aim to a) investigate wetting characteristics and phase-change heat transfer on multiscale textures; b) quantify electrowetting transport of condensates and transient heat transfer on the textured surface; c) develop numerical models of heat transfer on the textures. Reduced gravity flight tests will be performed to validate our analysis and ground tests. This innovative design from this program will enable a new generation of high-performance space integrated systems to operate at high-power densities without problems from temperature gradients, increased weight, or added complexity. Low-pressure-loss droplet transport on multiscale textures, combined with adaptable, low-power electrowetting, has the potential to significantly improve vapor chamber efficiency, increase its robustness and lifetime, and reduces the mass of space flight systems needed to support future human space exploration activities. The application of nanothermal materials and adaptive liquid transport technology based on electrowetting will lead to a major breakthrough in the thermal management field. The success of the proposed effort can considerably enhance thermal management in the future space explorations.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	One of the major applications of phase-change heat transfer at low-gravity conditions is the design and maintenance of robust and efficient thermal control systems for future space exploration and satellite applications. Heat pipes and vapor chambers based on phase-change heat transfer mechanisms have proven to be good candidates to dissipate heat in many space thermal-energy systems. However, a layer of capillary wick structure fully covering the heat pipe and vapor chamber inner walls creates additional thermal resistance because the effective thermal conductivity is much lower in saturated wicks. This thermal barrier can generate an unacceptable temperature rise, especially in high heat-flux and harsh-environment applications. To address this issue, Teledyne Scientific & Imaging Company (TS&I) proposes to develop an innovative wickless and electrowetting-controlled vapor chamber with bioinspired two-tier textures on the chamber wall. This innovative design will enable a new generation of high-performance space integrated systems to operate at high-power densities without problems from temperature gradients, increased weight, or added complexity. Low-pressure-loss droplet transport on multiscale textures, combined with adaptable, low-power electrowetting, has the potential to significantly improve vapor chamber efficiency, increase its robustness and lifetime, and reduce the mass of space flight systems needed to support future human space exploration activities. The application of nanothermal materials and adaptive liquid transport technology based on electrowetting will lead to a major breakthrough in the thermal management field. The success of the proposed effort can considerably enhance thermal management in the future space explorations.
Task Progress:	As a result of the increasing speed, expanding functionality, and decreasing size of microprocessors and integrated chips, heat dissipation power density has been rapidly growing over the past two decades. In response to the rapid advances in microelectronics, novel cooling technologies are needed to meet increasing cooling requirements. As a paradigm-shifting technique, electrowetting on dielectric (EWOD) uses electric potential to control the movement of a liquid droplet on a dielectric surface and has become one of the promising techniques for manipulating tiny amounts of liquids. Recently, due to its digital programmable and reconfigurable characteristics, electrowetting transport of liquid droplets has drawn significant interest in thermal management. The electrowetting effect is defined as the change in contact angle between an electrolyte and a solid surface as a result of an applied potential difference between them. In the presence of control electrodes specially patterned on the substrate surface, changes in the contact angle can give rise not only to continuous variations of the droplet shape, but also to droplet transport between the electrode units. For example, if a sufficiently high voltage is applied to an electrode unit overlapping a portion of the droplet sitting above, the generated driving force (i.e., gradient of the electrostatic energy) drives the droplet to align itself with the charged electrode. By turning on electrodes in sequence, droplets can be shuttled successively between any two units within the array, i.e., digital and programmable. While most other EWOD devices typically require sandwiching a liquid droplet between two parallel plates that contain opposing electrodes (i.e., parallel-plate configuration or dual-plate configuration), we have designed coplanar electrodes to actuate liquid droplets on a single open-plate, i.e., an open-plate or single-plate configuration. To actuate a droplet, the electrode units were used interchangeably as the activating and the ground electrodes. With a two-dimensional array of control electrodes patterned on a surface, cooling droplets can be transported along a programmable path to a hot spot without the need for mechanical moving parts. Compared with the parallel-plate arrangement, our single-plate design leads to relatively low flow friction, which arises only from the base plate, and therefore enables low-voltage (10-15VDC) EWOD operations. The simple design and fabrication procedures further enable the convenient integration of EWOD devices into an application package. The simultaneous and independent electrowetting manipulation of liquid droplets necessitates a multi-channel controller for the digital operation and programmable transport. We have designed and fabricated a multiple-channel DC/AC controller for the electrowetting system. The custom-built DC/AC voltage controller can provide -100V to 100V on 128 channels. Frequency can be changed by command from 1Hz to 1000Hz. The advantages of DC control are low power consumption and fabrication simplicity. However, the dielectric coating on the device surface may be subject to breakdown or permanent polarization when a high DC electric field is applied over an extended period. Alternatively we can apply AC control voltage, which usually consumes more electric power and is relatively more complicated. We use either a Complex Programmable Logic Device (CPLD) or a Microcontroller Unit (MCU) as a digital processing unit. A Switch Control Logic unit that contains mainly a comparator is used to secure the correct sequence of on/off operation and accurate voltages for the AC waveform. The Switches/Drivers function is implemented with a MOSFET device controlled by the output of the Switch Control Logic unit. The control output is fed back to the Switch Control Logic unit to ensure the output value is correct. The communication to the PC goes through a parallel interface for CPLD (or a serial

	interface for MCU). The DC/AC controller is designed to actively and adaptively control the electrowetting operation. Once programmed, the stand-alone controller can run without human intervention. In an open-plate configuration, we developed an analytical RC network model for estimating the EWOD driving force under AC condition and an energy-minimization-based model for DC actuation. According to the models, the electrode elements on the droplet's projected track are dynamically assigned as either driving electrodes or grounding electrodes in EWOD operations. With the driving voltage as low as 40 VAC, we successfully demonstrated two dimensional EWOD transport of a liquid droplet in any direction along any desired flow path on a surface. Besides, for DC actuation, electrodes were energized for only 250 ms and were always deactivated after the actuation when the intermittent deactivation protocol was used. Similar to the AC actuations, this DC pulse prevents irreparable damage to the chips that could occur if a permanent polarization was formed in the dielectric layer. The flexibility and stability of EWOD in combination with its cost effectiveness and portability make it an ideal platform to address a wide range of problems in chip cooling. During the first year of this program, we successfully developed an EWOD-based microfluidic technique for active and adaptive thermal management of on-chip hot spots. A two-dimensional array of control electrodes was patterned on the chip surface for EWOD operations. By applying DC or AC voltages with appropriate sequence and timing to the electrode units, we were able to transport microdroplets of tens of microliters along a programmable path. Without the need of external pumps and valves, the droplets were precisely delivered to the cooling targets. With EWOD-based microdroplet actuations, the hot spot surface was cooled down by more than 40ºC within 2 seconds. Furthermore, we demonstrated heat flux (7.6 W/cm2) cooling on a hot spot and the EWOD-induced internal circulation within the droplets led to a time-averaged Nusselt number of ~45.
Bibliography Type:	Description: (Last Updated: 05/11/2012)
Articles in Peer-reviewed Journals	Cheng J-T, Chen C-L. "Active thermal management of on-chip hot spots using EWOD-driven droplet microfluidics." Experiments in Fluids, Online First™, 28 April 2010. http://dx.doi.org/10.1007/s00348-010-0882-4 , Apr-2010